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NPS/003/002 – Technical Specification for 30V Tripping Battery and Charger Systems

1. Purpose

This document is the technical specification for 30V battery and charger systems for use by CE Electric UK.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
DSS/003/002	1	Technical Specification for 30V Battery and Charger Systems

2. Scope

This specification details the requirements for 30V battery and charger systems for use at distribution substations where DC supplies are necessary to operate protection relays or push-button operated trip coils.

3. Technical Requirements

Battery and charger systems or batteries for use without chargers shall be supplied as specified.

Batteries for use without chargers shall comprise primary cells and shall comply with the requirements in Appendix 1.

Battery and charger systems shall comply with the specification in the appropriate table in Appendix 2.

Note: Two types are specified: Type 1 - No Standing Load

Type 2 - Standing Load (load profile as described)

The equipment shall comply with the current editions of BS EN 60086, BS EN 60622, BS EN 6290-4 and ENATS 50-18 except where varied by this specification and associated appendices.

Batteries for use without chargers shall be entirely maintenance free but shall be provided with an inbuilt test facility as described in Appendix 1.

The technical specification of any system must be declared using the table in Appendix 3.

Where a standing load exists, information on the battery discharge profile is detailed in Appendix 2.

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4. References

4.1 External Documentation

Reference	Title
BS EN 60086	Primary Cells
BS EN 60622	Secondary cells and batteries containing alkaline or other non-acid electrolytes – sealed nickel cadmium prismatic rechargeable single cells
BS EN 6290-4	Lead-acid stationary cells and batteries. Specification for classifying valve regulated types.
ENATS 50-18	Design and application of ancillary equipment

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
Primary Cell	Source of electrical energy obtained by direct conversion of chemical energy, not designed to be charged by any other electrical source

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
V Gorny	Standards Engineer	V Gorny	05/10/2009

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	System Strategy	M Nicholson	21/10/2009

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
K Midgley	Standards Engineer	K Midgley	26/10/2009

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S.Johnson	DBD Administrator	S.Johnson	02/11/2009

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C.Holdsworth	Standards and Assessment Manager	C.Holdsworth	27/10/2009

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C.Morgan	Head of Procurement	C.Morgan	27/10/2009

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Appendix 1: Technical Specification

30V tripping battery without battery charger.

Battery Output Voltages	
Nominal voltage	30v
Minimum voltage after specified load duty	28v
Maximum open circuit voltage	38v
Load and Test duties	
Load pulse	6 amps for a duration of 150ms
Test pulse	6 amps for 200ms through a 5 ohm resistive load.
Load duty	Ten test pulses during commissioning. An average of three tripping operations and one test operation per year over the service life of the battery. No more than 10 operating pulses in any one year after commissioning.
Other Battery Design Factors	
Minimum service life	10 years
Standing load	None
Maintenance requirement	Maintenance free
Rated temperature	15°C
Temperature range	- 5°C, +40°C
Other Requirements	
Test facility	An inbuilt test facility shall be provided to apply the test pulse specified above. The output shall be easily read and the test facility shall automatically reset to prevent excessive discharge of the battery.
Housing	Wall mountable cabinet with external pre-drilled brackets and lockable doors. Approx dimensions :- 600(H) x 500(W) x 200(D) mm.
Cabinet earthing lug	10mm phosphor bronze or high tensile brass earthing stud.
DC output terminals and connectors	Shrouded, to accept 2.5 mm ² ring tongue terminals, for example types A, E in ENATS 50-18.
DC output fuses	2 x 6A fuses shall be provided for DC output and be mounted within the cabinet.
Secondary wiring	ENATS 50-18. Not less than 2.5 sq mm 7/ 0.67mm for battery circuits and 1.5 sq mm 30/0.25 for other circuits.
Battery earthing	The battery shall be unearthed.

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Battery disposal	An approved method must be provided.
Supporting documentation	Circuit diagram. General arrangement drawing. Descriptive details of cells and unit.
Labels	Fuselinks, instruments and switches shall be adequately labelled to indicate their function. A label is required on the facia to enable recording of commissioning and subsequent annual tests (to year 10). Labels shall be black on white.

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Appendix 2: Technical Specification for 30V battery and battery-charger system.

Battery Output Voltages and Load Duty	
Nominal voltage	30V
Maximum Float voltage	34V
Minimum voltage after specified discharge profile	28V
Maximum voltage under all conditions	37.2V
Battery standing load (Type 2)	1A
Battery discharge profile (Type 2)	1A / 10 hr
Other Battery Design Factors	
Standard	
Type	
Service life	10 years
Number of cells	
Battery Capacity	To meet the discharge profile indicated
Ageing factor	Typically 1.25 lead acid / 1.1 ni-cad
Minimum topping-up period	N/A
Rated temperature	15°C
Temperature range	-5°C, + 40°C
Mounting	Cubicle
Connectors	Shrouded
Isolation facilities required	From charger and distribution board
Battery earthing	The battery shall be unearthed
Automatic Battery Disconnect	Automatic battery disconnect facility shall be provided (if required, depending on battery type to prevent damage). Battery to be reconnected on restoration of charger supply.
Charger Supply	
AC Supply	230v AC single phase
Nominal supply frequency	50Hz
Supply frequency range	47-52Hz
Charger	
Type	Constant voltage, two-stage, temperature compensated.
Ambient temperature range	-5°C, + 40°C
Maximum Float Voltage	34V
Maximum voltage under all conditions	37.2V
Boost charge selection	Boost selector switch to be mounted behind fascia door.

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Voltage adjustment facility	Float and boost voltages should be adjustable however such controls should not be on the front facia.
Rated charging current	To cover standing load as specified plus adequate charging current
Normal float charge	50% rated load at rated input voltage.
Variation in charging voltage as input voltage varies over range of -10% + 6% of rated value:	
Across load range 0 – 10%	Maximum 2% variation in output voltage
Across load range 10% - Full Load	Maximum 1% variation in output voltage
Variation in current limit over the specified voltage range of the battery.	Maximum 2% of the nominal setting
Frequency range over which charger performance should be met.	47-52Hz
Max earth current with one pole of battery earthed (at maximum battery voltage)	5mA
Transformer screening	Earth metal screen between primary and secondary windings.
Secondary wiring	ENATS 50-18 Not less than 2.5 sq mm 7/0.67mm for battery circuits and 1.5 sq mm 30/0.25 for other circuits.
System Monitoring	
High voltage alarm	Local indication plus spare contact
Low voltage alarm	Local indication plus spare contact
Charger fail alarm	Local indication plus spare contact
Battery earth fault alarm	Local indication plus spare contact
Battery high resistance or open circuit	Local indication plus spare contact
Battery Disconnect alarm	Local indication plus spare contact
Monitoring alarm protection	2A HRC fuse
Battery earth fault alarm test facility	
Test resistor value	10% below sensitivity of the earth fault relay
Operating switch	A "return-to-off" on/off switch
Discharge Test Facility	
Discharge resistor	To give one hour discharge rate
Operating method	Via a "return-to-off" on/off switch, and contactor.
Warning label	"This switch shall not be operated for longer than 10 seconds"
Instruments	
Battery output voltmeter	Push-button or self-return-switch operated
Charger output ammeter	

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Cubicles and Battery Stands	
General arrangement	Charger to be housed in a cabinet with lockable hinged facia. Batteries to be housed on tiered shelves in the same cabinet behind lockable doors. Approx dimensions :- 800(H) x 600(W) x 300(D) mm.
Design life	40 years
Cubicle cable entry	Bottom
Cubicle Colour	Light grey
Exposed conductors	Shrouded
Doors	Lockable
Distribution Board	
Number of ways	Min 2
Rating of ways	16A
Labels	
Fuses	
Type	Cartridge type: BS88

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Appendix 3 - SELF CERTIFICATION CONFORMANCE DECLARATION

30V Battery and Charger Systems required to be supplied against this specification shall comply with the latest issues of the relevant ENATS, British and International Standards specified. The following tables are intended to amplify and/or clarify the requirements of elements of these Standards but do not preclude meeting all requirements of the standards. .

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary.
- When any other code is entered the reason for non-conformance shall be entered.
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

Details of the Cable Type (Voltage, Conductor Type and Size)

Name:

Signature:

Date:

NOTE:

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TECHNICAL SPECIFICATION FOR 30V TRIPPING BATTERY				
		Clause / Requirements	Conformance Code	Remarks / Comments
	System Output Characteristics			
	Nominal voltage	30V		
	Minimum voltage after specified discharge profile	28V		
	Maximum charger voltage under all conditions	38V		
	Maximum open circuit voltage	State		
	Battery			
	Standard	State (BS/IEC?)		
	Type of cells	State		
	Service life	10 Years		
	Aging factor applied	State		
	Number of cells in battery	State		
	Float voltage per cell	State (V)		
	Rated temperature	15 °C		
	Temperature range	-5 / +40 °C		
	Cell mounting: Cubicle? Tiered?	State		
	Connectors	Shrouded		
	Isolatable from dist board	Required		
	Battery Earthing	Unearthed		
	Test Facility	Required		

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TECHNICAL SPECIFICATION FOR 30V TRIPPING BATTERY				
		Clause / Requirements	Conformance Code	Remarks / Comments
Discharge Test Facility				
	Indication	1h d/charge rate		
	Operating Method	"return to off" type switch		
	Warning label	Required		
Cubicles and Battery Stands				
	Design life	20 Years		
	Cubicle cable entry position	Bottom		
	Cubicle Colour	Light Grey		
	Exposed Conductors	Shrouded		
	Doors	Lockable		
	Dimensions (approx 600 x 500 x 200mm)			
Distribution Board				
	Number of ways	Minimum 2		
	Rating of ways	16 A		
Fuses				
	Type	BS88		

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TECHNICAL SPECIFICATION FOR 30V BATTERY AND CHARGER SYSTEMS				
		Clause / Requirements	Conformance Code	Remarks / Comments
System Output Characteristics				
	Nominal voltage	30V		
	Float voltage (max acceptable)	34V		
	Minimum voltage after specified discharge profile	28V		
	Maximum charger voltage under all conditions	37.2V		
	Maximum open circuit voltage	State		
Battery				
	Standard	State (BS/IEC?)		
	Type of cells	State		
	Service life	10Years		
	Aging factor applied	1.25 lead acid 1.1 ni-cad		
	Minimum topping-up period	N/A		
	Number of cells in battery	State		
	Float voltage per cell	State (V)		
	Rated temperature	15 °C		
	Temperature range	-5 / +40 °C		
	Cell mounting: Cubicle? Tiered?	State		
	Connectors	Shrouded		
	Isolatable from both dist board and charger	Required		
	Battery Earthing	Unearthed		
	Test Facility	Required		
	Auto Battery Disconnect	State		

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TECHNICAL SPECIFICATION FOR 30V BATTERY AND CHARGER SYSTEMS				
		Clause / Requirements	Conformance Code	Remarks / Comments
Charger Supply				
	AC Supply	230V / 1 Phase		
	Nominal supply frequency	50Hz		
	Supply frequency range	47-52Hz		
Charger				
	Type	Constant Voltage		
	Ambient temperature range	-5 / +40 °C		
	Maximum Float Voltage	34 V		
	Maximum Voltage	37.2V		
	Voltage adjustment facility (float and boost - located behind fascia).	Required		
	Rated charging current	Charging current + 1A		
	Normal float charge	0.5A at rated input voltage		
	Variation in charging voltage as input voltage varies over range of -10% + 6% of rated value:			
	Across load range 0 – 10%	Max 2 %		
	Across load range 10% - Full Load	Max 1%		
	Variation in current limit over the specified voltage range of the battery.	Max 2% of nominal		
	Frequency range over which charger performance should be met.	47-52Hz		
	Max earth current with one pole of battery earthed (at maximum battery voltage)	5 mA		
	Transformer screened as per specification	Required		
	Secondary Wiring	ENATS 50-18		

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TECHNICAL SPECIFICATION FOR 30V BATTERY AND CHARGER SYSTEMS			
	Clause / Requirements	Conformance Code	Remarks / Comments
System Monitoring			
High voltage alarm setting (adjustable)	Local+Contact *		
Low voltage alarm setting (adjustable)	Local+Contact *		
Charger fail alarm	Local+Contact *		
Battery earth fault alarm	Local+Contact *		
Battery high resistance or open circuit	Local+Contact *		
Facility for remote alarm common alarm	Local+Contact *		
Alarm circuit fuse rating	2 A		
Battery Damage (Auto Disconnect / Restore)	Auto Disconnect		
Battery earth fault alarm test facility			
Operating Method	"return to off" type switch		
Test Resistor (% below sensitivity of E/F relay)	10%		
Instruments			
Battery output voltmeter	Push or self reset switch		
Charger output ammeter	Required		
Cubicles and Battery Stands			
Design life	40 Years		
Cubicle cable entry position	Bottom		
Cubicle Colour	Light Grey		
Exposed Conductors	Shrouded		
Doors	Lockable		
Dimensions (approx 800 x 600 x 300mm)			
Distribution Board			
Number of ways	Minimum 2		
Rating of ways	16 A		
Fuses			
Type	BS88		

* - Local indication + "spare" contact available for remote alarm